

COMP3011 Computer Graphics

Spring 2024

Assessment 3

Report Sheet (v2)

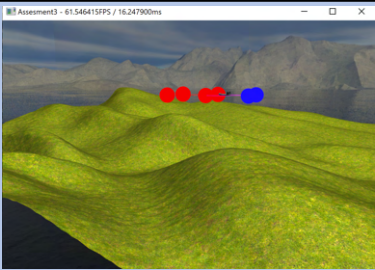
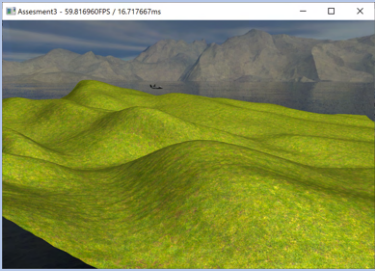
This table will help you prepare for your demo. **Submit this report to Moodle.**

Student Name: Nathan Burgess

Student ID: 20363169

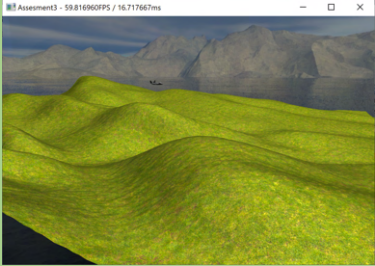
Username: psynb7

I agree for my code & report to be published, with my name, to future students as an example (yes/no): yes

Introduction		
<i>Please explain why you implemented this scene</i>	<i>Describe your inspirations</i>	<i>Provide a general description of the scene.</i>
It gave a good opportunity for nice animations, open lighting from the sun and the ability to mess around with generating heightmap terrain.	<i>I'm a fan of planes so thought it would be nice to have something.</i>	<i>Grassy hills with a jet flying above and some scattered trees.</i>
TR 2 – 3D Modelling		
Object 1 - procedurally generated		
<i>Please give a screenshot</i>	<i>reference specific code (filename and line)</i>	<i>Description of object</i>
 	<i>Terrain.cpp CompleteObject.cpp - `constructHeightMapTerrain` line 9 110 (flat plane but unused)</i>	Rolling grassy hills generated using a heightmap. Uses indices to save on VRam

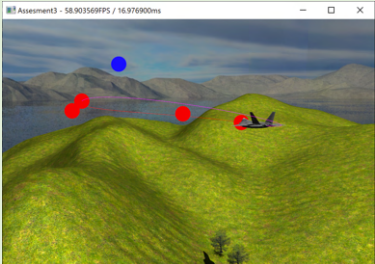
Object 2 - OBJ parser		
<i>Please give a screenshot</i>	<i>Provide the URL for the OBJ file(s) you submitted</i>	<i>Describe what you did to extend your OBJ parser from CW2.</i>
	https://sketchfab.com/3d-models/f-22-raptor-update-d-fighter-jet-free-7bfc05d3916b454da4960fc17c093874 <i>Tree from CW2</i> \Assessment3\Assessment3\Assessment3\objs	Normal Coordinates un-reversed UV coordinates (flip with stbi instead) Ns, Ka, Kd and Ks for each material to allow for specific lighting. Modified to use std::optional and std::string to be nicer to use.

TR 3 – 3D Transformations

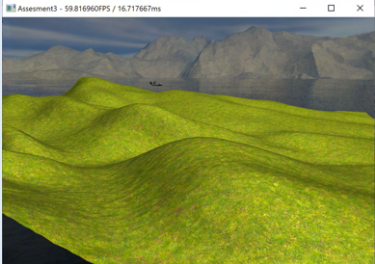
Object 1 - procedurally generated		
<i>Please give a screenshot of transformed object</i>	<i>reference specific code (filename and line)</i>	<i>Description of transformations</i>
	Terrain.cpp - line28	A y shift is used to move the terrain up to be centred on 0,0 more.
Object 2 - OBJ parser		
<i>Please give a screenshot of transformed object</i>	<i>reference specific code (filename and line)</i>	<i>Description of transformations</i>
	Assessment3.cpp 156 - 169 CompleteObject.cpp 76, 80, 21	Plane animated using translate + rotate to follow a curve. Plane + trees scaled to fit the scene size. Trees transformed to be placed correctly.

		
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TR 3 – Animation

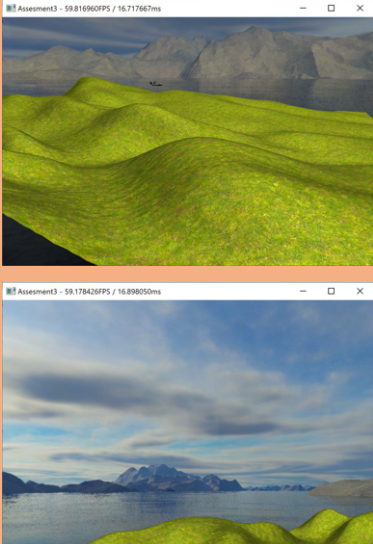
<i>Please give a screenshot of animated object</i>	<i>reference specific code (filename and line)</i>	<i>Description of animation</i>
	<i>F22.cpp - Full file (specifics on line 24)</i> <i>CompleteObject - 21.</i>	<i>Plane flies around interpolated bezier curve to go around an oval in 3D space and follow the tangent of the curve.</i>

TR 4 – Camera

<i>Please give a screenshot</i>	<i>reference specific code (filename and line)</i>	<i>Description of camera</i>
	<i>camera.cpp</i>	Flythrough Camera

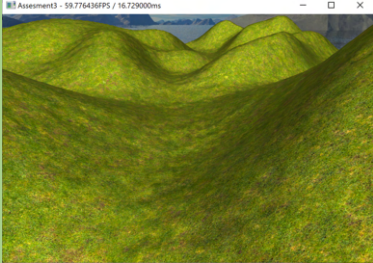
TR 5 – Texture

<i>Object 1 - procedurally generated</i>		
<i>Please give a screenshot</i>	<i>reference specific code (filename and line)</i>	<i>Description of texture</i>

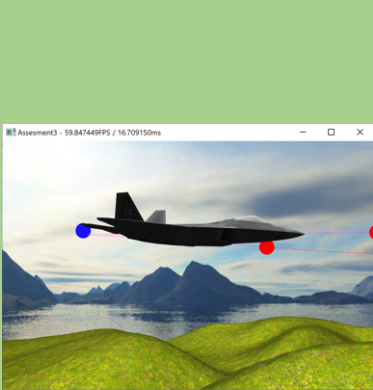
	<i>Texture.cpp, - 3 mipMaps - 37, Terrain.cpp 47</i> <i>Skybox.frag .vert and .cpp</i> <i>Texture.cpp 47</i>	<i>Uses mipmaps UV's calculated to set resolution to tessellate seamlessly and still appear high res.</i> <i>Skybox uses cube mapping.</i>
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TR 6 – Lighting

Object 1 - procedurally generated

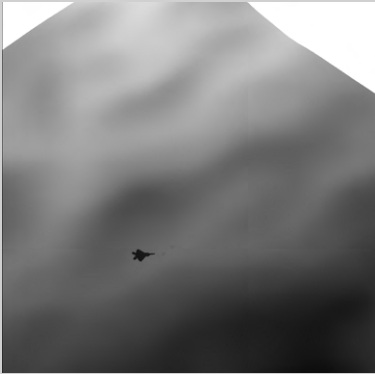
<i>Please give a screenshot</i>	<i>reference specific code (filename and line)</i>	<i>Description of light(s) on object</i>
	Phong.frag, vert	Blinn-phong lighting Directional (spot and positional are implemented but un-used)

Object 2 - OBJ parser

<i>Please give a screenshot</i>	<i>reference specific code (filename and line)</i>	<i>Description of light(s) on object</i>
	Phong.frag, vert	Blinn-phong lighting Directional spot and positional are implemented but un-used)

TR 7 - Shadow

Object 1 - procedurally generated

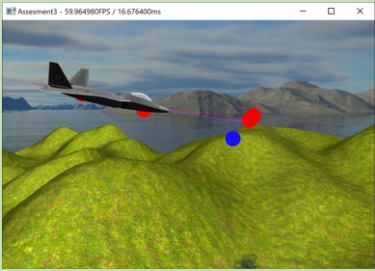
<i>Please give a screenshot</i>	<i>reference specific code (filename and line)</i>	<i>Description of shadow on object</i>
	Phong.frag -162 ShadowRenderin g.cpp shadow.cpp	Shadows use PCF to increase resolution.

Object 2 - OBJ parser

<i>Please give a screenshot</i>	<i>reference specific code (filename and line)</i>	<i>Description of shadow on object</i>
  	Phong.frag -162 ShadowRenderin g.cpp shadow.cpp shadow.frag,vert	Shadows use PCF to increase resolution. Pass through texture transparency to improve shadow quality for tree.

TR 8 - Interactive object

<i>Please give a screenshot</i>	<i>reference specific code (filename and line)</i>	<i>Description of interactive object</i>
NAN	NAN	NAN

TR 9 – Curves		
<i>Please give a screenshot</i>	<i>reference specific code (filename and line)</i>	<i>Description of curve(s)</i>
	<i>Castelijau.cpp (wholeFile)</i> <i>F22.cpp (wholeFile)</i>	<i>Two bezier curves in a 3D plane used for object animation.</i> <i>Tangent of line is calculated to apply rotation to the animated object.</i>
TR 10 – Transparency		
<i>Please give a screenshot</i>	<i>reference specific code (filename and line)</i>	<i>Description of transparency</i>
	<i>Phong.frag 43,44</i>	<i>If transparency is present in a texel we drop the fragment which uses that.</i>
R&D		
Please provide details of any research and development you conducted, as additional technique(s) not in the lecture notes.		
<i>Please give a screenshot</i>	<i>reference specific code (filename and line)</i>	<i>Description of Research including websites, articles, references, etc.</i>
	<i>Phong.frag 55 - 63</i> <i>Terrain.cpp 89, 9 texture.cpp 3, 79</i> <i>Skybox.cpp texture.cpp 47</i> <i>F22.cpp, castelijai.cpp, 34</i> <i>Phong.frag -162</i>	<i>Blinn-phong lighting</i> https://learnopengl.com/Advanced-Lighting/Advanced-Lighting <i>Heightmaps</i> https://learnopengl.com/Guest-Articles/2021/Tessellation/Height-map <i>Skybox + cubemap</i> <i>Bezier curve tangents for use as animation</i> https://stackoverflow.com/questions/25453159/getting-consistent-normals-from-a-3d-cubic-bezier-path <i>PCF (improve shadow quality)</i>

	Assessment3.cpp 280 - 282, obj.cpp 28 Terrain.cpp - 50	https://learnopengl.com/Advanced-Lighting/Shadows/Shadow-Mapping <i>Alternate light properties</i> https://learnopengl.com/Lighting/Materials <i>Indices</i> https://learnopengl.com/Guest-Articles/2021/Tessellation/Height-map
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